NETWORKING – DAY 02

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THEORY

1. What is a computer network?

A computer network is a group of interconnected computers and devices that communicate with each other and share data, resources, and services using communication protocols.

2. What is the difference between a client and a server?

A client is a device or application that requests data or services. A server is a computer or system that receives requests and provides data, resources, or services to clients.

3. What is a LAN?

LAN stands for Local Area Network. It is a network that connects computers and devices within a small geographical area such as a home, office, school, or laboratory.

4. What is a WAN?

WAN stands for Wide Area Network. It connects networks over a large geographical area such as cities, countries, or continents. The Internet is the largest example of a WAN.

5. What is the purpose of a switch?

A switch connects multiple devices within a local network and forwards data to the correct device using MAC addresses. It helps devices communicate efficiently inside a LAN.

6. What is the purpose of a router?

A router connects different networks and forwards data packets between them using IP addresses. It is commonly used to connect a local network to the Internet.

7. What is a firewall?

A firewall is a security system that monitors and controls incoming and outgoing network traffic according to predefined security rules. It helps prevent unauthorized access to a network or computer.

8. What is an IP address?

An IP address is a logical address assigned to a device on a network. It is used to identify the device and enable communication between devices. IPv4 is commonly written in four decimal numbers, such as 192.168.0.101.

9. What is a MAC address?

A MAC address is a unique hardware address associated with a network interface. It is used to identify devices at the data link layer of a network and is usually written in hexadecimal format.

10. What is a default gateway?

A default gateway is the device, usually a router, through which a computer sends traffic destined for another network, such as the Internet.

11. What is DNS?

DNS stands for Domain Name System. It translates human-readable domain names, such as google.com, into IP addresses so that computers can locate the required servers.

12. Why is networking knowledge important for cybersecurity?

Networking knowledge is important for cybersecurity because attacks, data transmission, authentication, and security controls all depend on networks. Understanding IP addresses, ports, protocols, DNS, routing, and traffic helps security professionals detect, analyze, and prevent attacks.

MY NETWORK INFORMATION

13. Write your computer's private IPv4 address.

- 192.168.0.101

14. Write your default gateway.

- 192.168.0.1

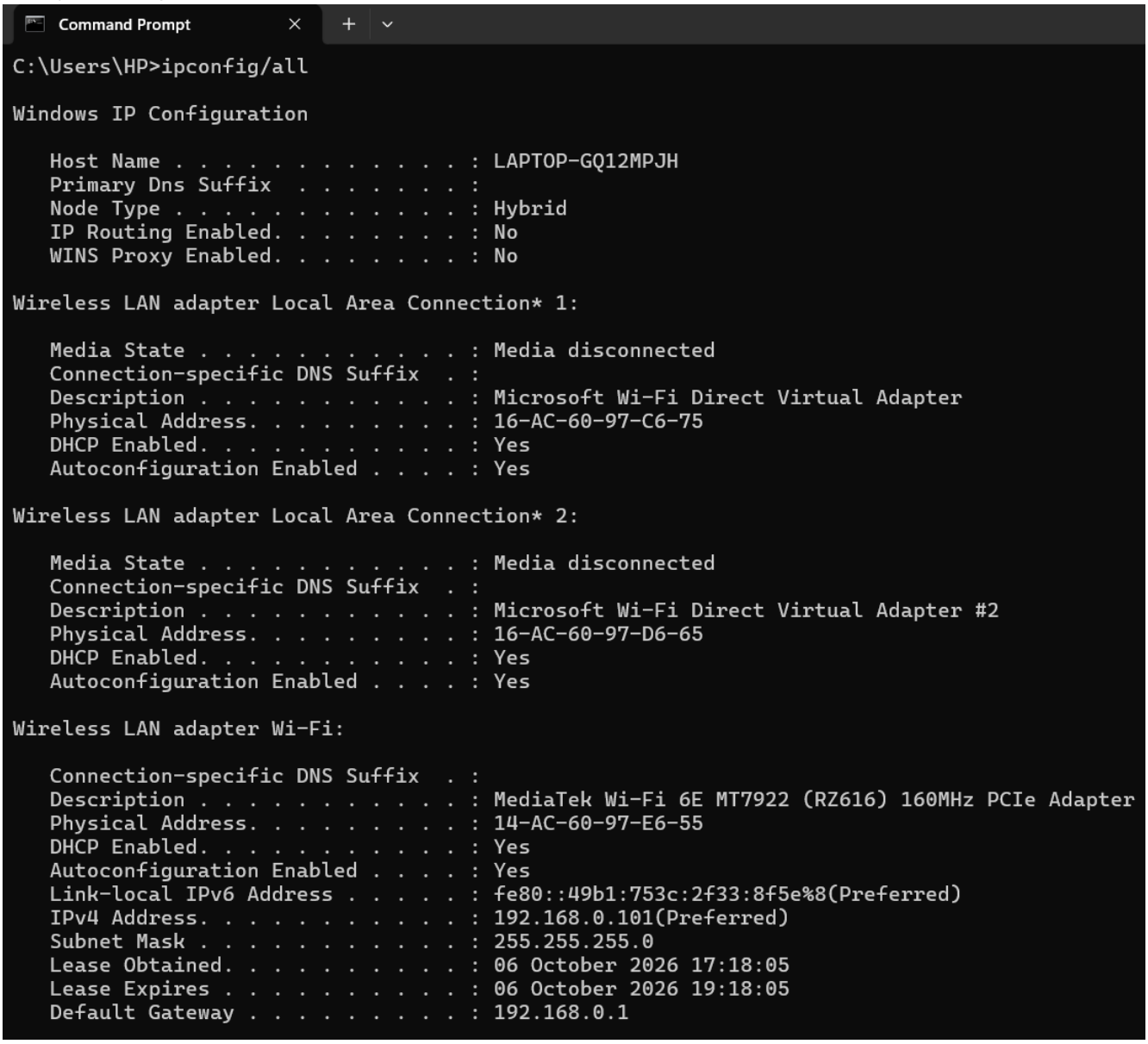
15.Write your DNS server.

- 192.168.0.1

❖ SCREENSHOTS (3 marks)

Attach screenshots of: ipconfig, ping, tracert

1) ipconfig



```
   DHCP Server . . . . . . . . . . . : 192.168.0.1
   DHCPv6 IAID . . . . . . . . . . . : 85240928
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-32-46-78-B6-48-EA-62-93-48-CB
   DNS Servers . . . . . . . . . . . : 192.168.0.1
   NetBIOS over Tcpip. . . . . . . . : Enabled

Ethernet adapter Ethernet:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Realtek Gaming GbE Family Controller
   Physical Address. . . . . . . . . : 48-EA-62-93-48-CB
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

C:\Users\HP>
```

2) ping

```
C:\Users\HP>ping google.com

Pinging google.com [142.251.106.113] with 32 bytes of data:
Reply from 142.251.106.113: bytes=32 time=65ms TTL=115
Reply from 142.251.106.113: bytes=32 time=12ms TTL=115
Reply from 142.251.106.113: bytes=32 time=35ms TTL=115
Reply from 142.251.106.113: bytes=32 time=11ms TTL=115

Ping statistics for 142.251.106.113:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 11ms, Maximum = 65ms, Average = 30ms
```

3) tracert

```
C:\Users\HP> tracert google.com

Tracing route to google.com [142.251.106.113]
over a maximum of 30 hops:

  1    27 ms     6 ms     5 ms  192.168.0.1
  2    24 ms     8 ms     8 ms  103.252.4.94.threesainfoway.net [103.252.4.94]
  3     *        *        *     Request timed out.
  4    19 ms     8 ms     8 ms  142.250.165.170
  5    33 ms    21 ms    34 ms  cm-in-f113.1e100.net [142.251.106.113]

Trace complete.
```